

Methods (CIFAR-10/WRN-28-10)	Acc $\uparrow$	NLL $\downarrow$	ECE $\downarrow$	cAcc $\uparrow$	cNLL $\downarrow$	cECE $\downarrow$	FLOPs $\downarrow$	Epochs $\downarrow$
Independent Model*	96.0	0.159	0.023	76.1	1.050	0.153	3.6e17	200
Monte Carlo Dropout*	95.9	0.160	0.024	68.8	1.270	0.166	1.00x	200
TreeNet (M=3)*	95.9	0.258	0.018	75.5	0.969	0.137	1.52x	250
SSE (M=5)	96.3	0.131	0.015	76.0	1.060	0.121	1.00x	200
FGE (M=12)	96.3	0.126	0.015	75.4	1.157	0.122	1.00x	200
PAT (M=6) (AR + 1C)	96.48	0.113	0.005	76.23	0.972	0.081	0.85x	200
BatchEnsemble (M=4)*	96.2	0.143	0.021	77.5	1.020	0.129	4.40x	250
MIMO (M=3)*	96.4	0.123	0.010	76.6	0.927	0.112	4.00x	250
EDST (M=7)*	96.4	0.127	0.012	76.7	0.880	0.100	0.57x	850
DST (M=3)*	96.4	0.124	0.011	77.6	0.840	0.090	1.01x	750
Dense Ensemble (M=4)*	96.6	0.114	0.010	77.9	0.810	0.087	1.00x	800
Methods (CIFAR-100/WRN-28-10)	Acc $\uparrow$	NLL $\downarrow$	ECE $\downarrow$	cAcc $\uparrow$	cNLL $\downarrow$	cECE $\downarrow$	FLOPs $\downarrow$	Epochs $\downarrow$
Independent Model*	79.8	0.875	0.086	51.4	2.700	0.239	3.6e17	200
Monte Carlo Dropout*	79.6	0.830	0.050	42.6	2.900	0.202	1.00x	200
TreeNet (M=3)*	80.8	0.777	0.047	53.5	2.295	0.176	1.52x	250
SSE (M=5)	82.1	0.661	0.040	52.2	2.595	0.145	1.00x	200
FGE (M=12)	82.3	0.653	0.038	51.7	2.638	0.137	1.00x	200
PAT (M=6) (AR + 1C)	82.67	0.634	0.013	52.70	2.487	0.131	0.85x	200
BatchEnsemble (M=4)*	81.5	0.740	0.056	54.1	2.490	0.191	4.40x	250
MIMO (M=3)*	82.0	0.690	0.022	53.7	2.284	0.129	4.00x	250
EDST (M=7)*	82.6	0.653	0.036	52.7	2.410	0.170	0.57x	850
DST (M=3)*	82.8	0.633	0.026	54.3	2.280	0.140	1.01x	750
Dense Ensemble (M=4)*	82.7	0.666	0.021	54.1	2.270	0.138	1.00x	800

Table 2: Results for ensembles of WideResNet-28-10 models on both CIFAR-10 and CIFAR-100. Methods with * denote results obtained from (??). Best low-cost ensemble results are **bold**. cAcc, cNLL, and cECE correspond to corrupted test sets. We report the mean values over 10 runs for PAT.

methods and are competitive with methods that train for significantly longer. Our approach produces well calibrated, robust and diverse ensembles with excellent performance on out of distribution corrupted datasets.

Conclusions

The Prune and Tune Ensemble algorithm is flexible and enables the creation of diverse and accurate ensembles at a significantly reduced computational cost. We do this by 1) training a single large parent network, 2) creating child networks by copying the parent and significantly pruning them using random or anti-random sampling strategies, and 3) fine tuning each of the child networks for a small number of training epochs.

With the experiments introduced here, Prune and Tune Ensembles outperform low-cost ensembling approaches on benchmark image classification datasets with a variety of popular deep neural network architectures. Our approach achieves comparable accuracy to ensembles that are trained on 4-5x more data. Prune and Tune Ensembles not only improve upon the training-time/accuracy trade-off, but also reduce memory and computational cost thanks to the use of sparse child networks.

Prune and Tune Ensembles can offer a new lens through which we can analyze generalization and diversity among subcomponents of deep neural networks. We hope to fur-

ther explore larger scale benchmarks, anti-random pruning methodologies and applications to new problem domains in future work.

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